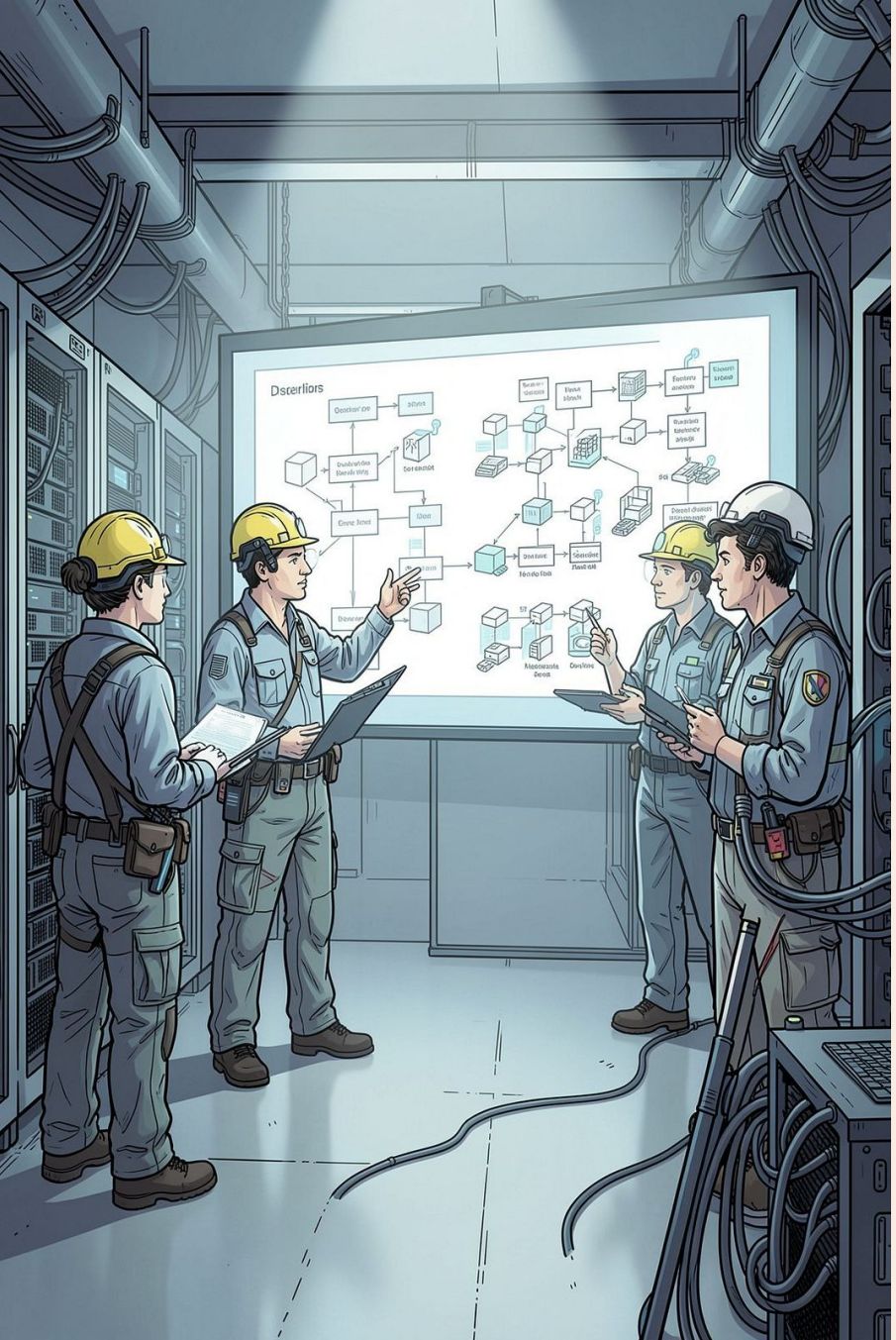

Lumen Academy — Database Schema & Architecture

Assessment engine and LMS built for NEET, JEE Main, and JEE Advanced. PostgreSQL 16 on Supabase, Prisma v6.14.0. 36 Prisma models across 11 domain modules. Security-first design: server-controlled clocks, IDOR-proof authorization, and zero answer leakage.

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Core Architectural Principles — 4 Non-Negotiable Schema Laws

Zero-Leak Answer Keys

Fields like `isCorrect` and `numericalAnswer` live only on server-side snapshots and are never selected or returned before attempt submission.

Server-Controlled Clocks

`startedAt`, `deadlineAt`, and sectional `lockedAt` are authoritative on server time — client clocks are untrusted.

IDOR-Proof Authorization

Every student row ties to `userId` or to a parent entity owned by that `userId` to prevent access-by-id guessing.

Immutable Result Freeze

`AttemptQuestion` snapshots question stem, options, marking rules, and answer keys at submit time — edits never touch historical scores.

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11 Domain Modules — 36 Prisma Models

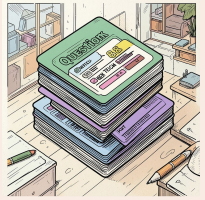
Concise module list and representative Prisma models for each domain. This structure guides ownership, RBAC boundaries, and rollup responsibilities for analytics and grading.

01 — Identity & Security

`User`, `StudentProfile`, `UserPreference`, `AuditLog`

02 — Syllabus Taxonomy

`Subject`, `Unit`, `Topic`



03 — Question Bank

`Question`, `Option` – supports 5 interaction formats

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Identity & Profile Architecture

Primary user model maps to Supabase auth; optional fields for email/phone and role-based defaults. StudentProfile feeds the AIR rank predictor via categorical inputs.

AIR Rank Inputs

`StudentProfile.category` (GENERAL, EWS, OBC_NCL, SC, ST) and `StudentProfile.state` — used to calculate home-state quotas and cutoffs.

Audit & Compliance

`AuditLog` stores hashed IP and privileged-action records for forensic review and compliance.

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Question Bank — 5 Interaction Formats & Review Lifecycle

SINGLE_CORRECT

Standard 4-option MCQs used for NEET/JEE Main.

MULTIPLE_CORRECT

Multiple answers with partial marking contracts for advanced exams.

NUMERICAL

Integer/decimal inputs graded with `numericalTolerance` metadata.

MATCHING

Left/right matrix mappings with validation rules and UI hints.

ASSERTION_REASON

Paired statements with combined correctness logic and partial scoring.

Editorial pipeline: DRAFT → PENDING_REVIEW → PUBLISHED → RETIRED with human critic paths and versioning.

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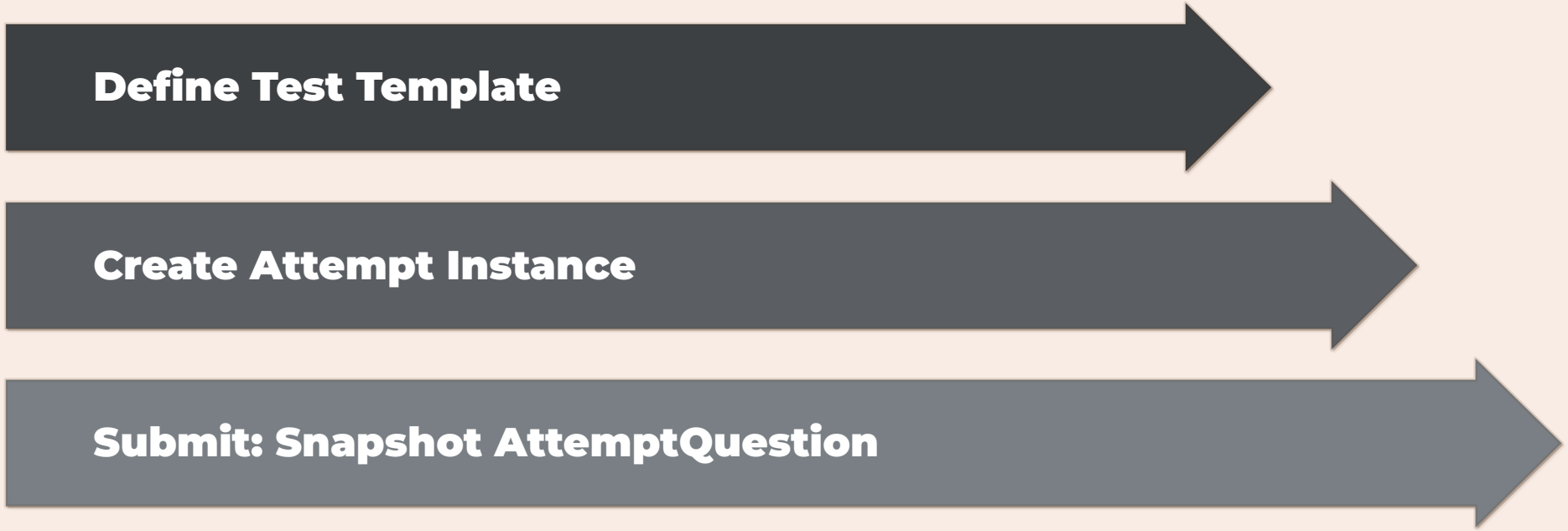

Separation of template and instance preserves templates for reuse while attempts hold authoritative, immutable snapshots used for grading and audit.

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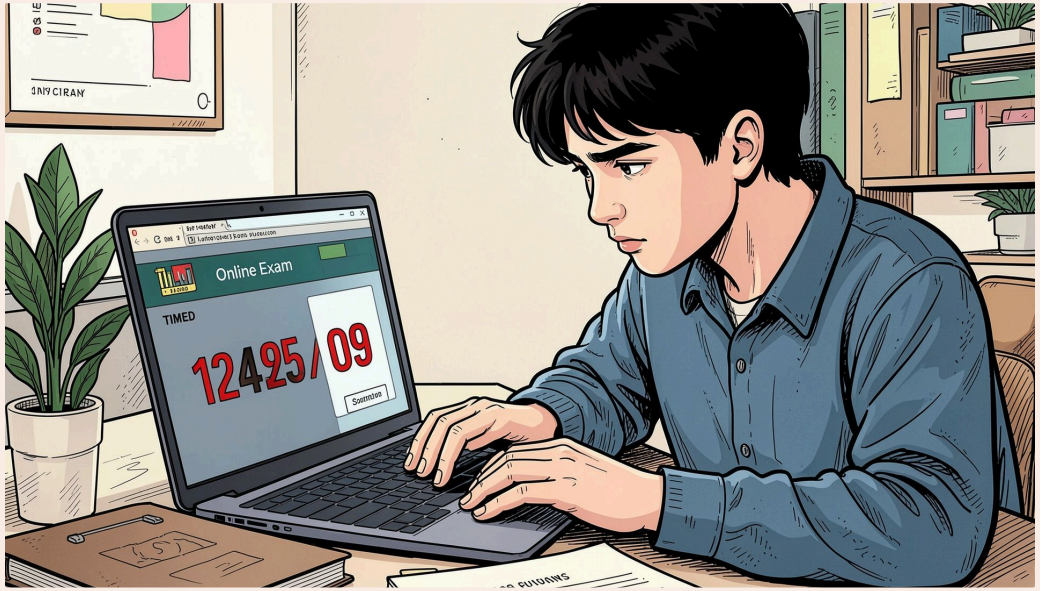

Attempt Engine — Sectional Timing & Autosave

- **Sectional Clock Enforcement**: Independent `AttemptSection.lockedAt` values let sections lock separately (required for JEE sectional rules).
- **Autosave Resilience**: `AttemptQuestion` uses upsert semantics keyed by `(attemptId, questionId)` to support intermittent connectivity.
- **Scoring**: Server computes marks at submission time using snapshotted rules; no client-side scoring trusted.

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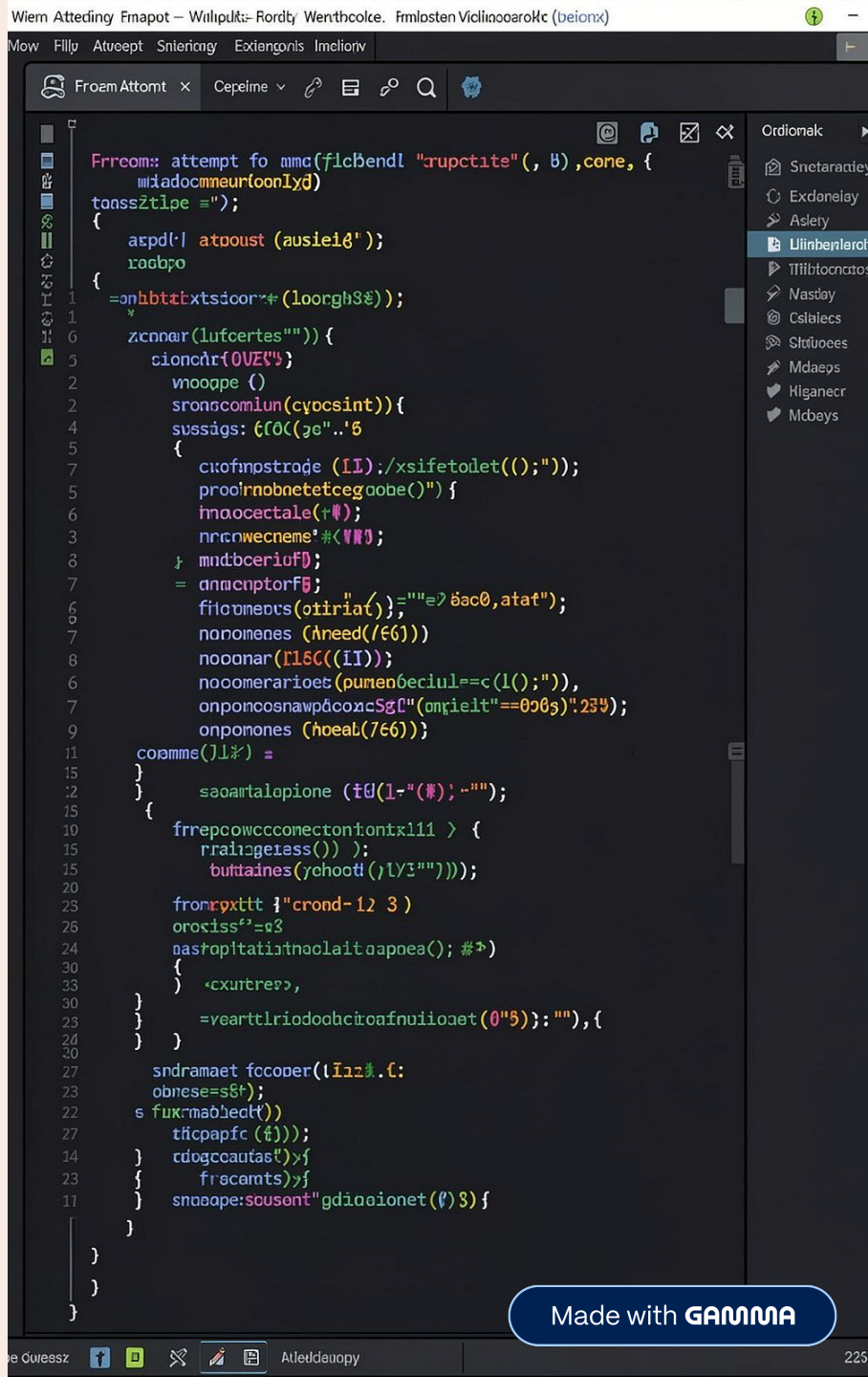
Law 4 — Immutable Snapshotting (Example)

```
{
  "attemptId": "att-9901",
  "questionId": "q-4412",
  "contentSnapshot": { "stemEn": "What is unit vector magnitude?" },
  "correctOptionIdsSnapshot": ["opt-a-uuid"],
  "marksSnapshot": 4,
  "negativeMarksSnapshot": 1,
  "marksAwarded": 4,
  "isCorrect": true
}
```

Business value: Corrections to live Question records never retroactively change historical student scores or transcripts.



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LMS, Gamification & Productivity Tools

Integrated content and retention features designed to increase daily active use and learning efficacy.

Lesson Video

Timestamps, bookmarks, and watch-percentage tracking (`LessonVideo`, `VideoTimestamp`, `LessonProgress`).

Gamification

`Streak`, `Badge`, `DailyGoal`, push `Notification` queue.

Productivity Suite

`PomodoroSession`, `StudyPlan` (AI-generated timetables), `Flashcard`, `RevisionNote`.

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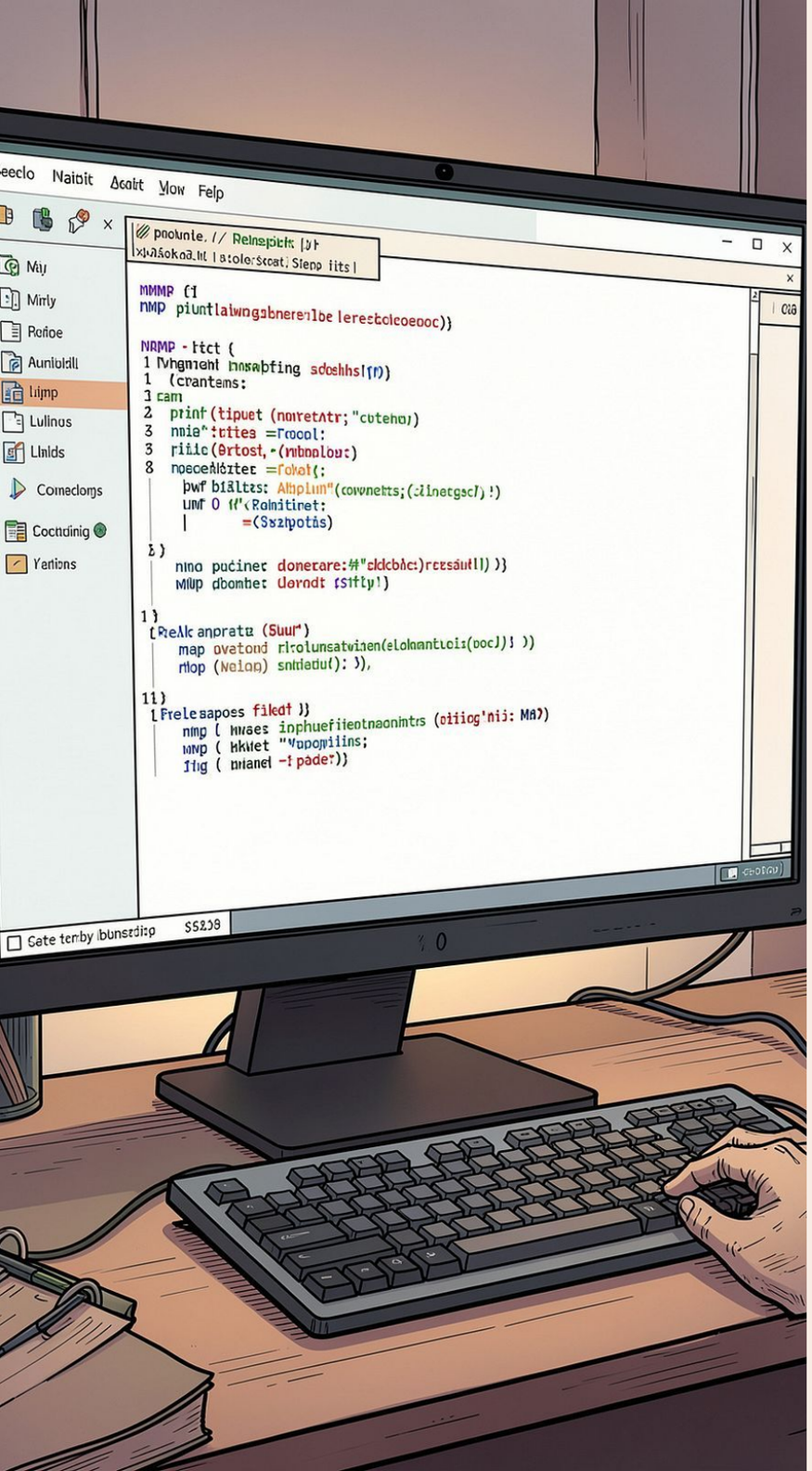

Developer Operations & Production Status

CLI and CI workflows ensure schema validity, type safety, and deployable artifacts.

DB Validation

`npm run db:validate` – schema syntax and relation checks.

Client Generation

`npm run db:generate` – generate type-safe `@prisma/client`.

Model Verification

`npm run db:verify-all` – populate and sanity-check all 36 models.

Studio & Typecheck

`npm run db:studio`; `npm run typecheck` passed clean on PostgreSQL 16 (Supabase).

Production note: All models validated and deployed with zero TypeScript errors. Follow-up items: formalize backup/restore SLAs, run periodic snapshot audits, and codify emergency rollback procedures.

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